Project Eco

M1: Approach | Green Home Score and Retrofit Recommendation Engine

01 Problem statement and understanding

Homeowners often want to improve the energy efficiency of their homes but are unsure which changes to make first. Lloyds Problem Statement 2 asks for a system that evaluates a home, identifies poor performance, recommends suitable improvements, and estimates potential energy savings, carbon reduction and property-value impact.

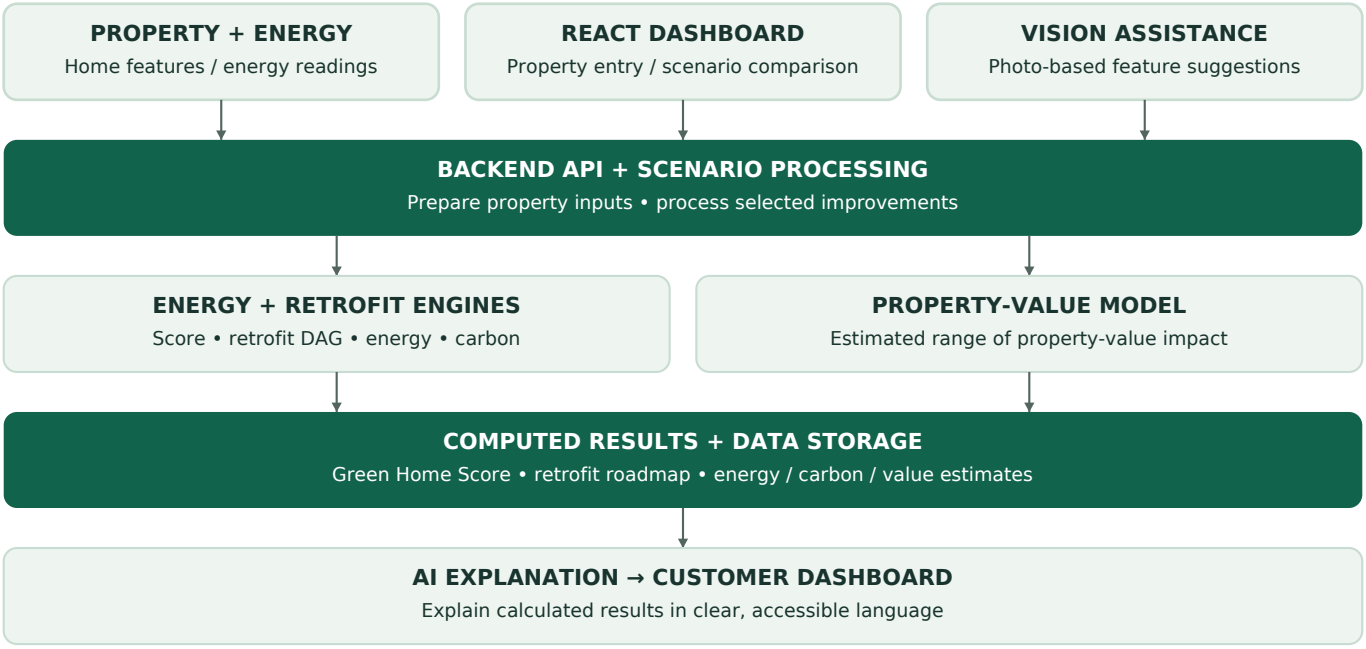
Our understanding is that recommendations must reflect the property and the interaction between improvements. Insulation reduces heating demand, heat pumps may increase electricity use while reducing gas, and solar panels reduce grid imports. The system must therefore compare realistic before-and-after scenarios and explain its results clearly.

02 Idea / abstract

Project Eco (EcoRetrofit AI Engine) is a proposed AI-enabled platform for homeowners and mortgage advisors. It combines an explainable Green Home Score, a prioritised retrofit roadmap, property-value modelling and a conversational what-if assistant in one dashboard.

Users will provide property details and energy readings, review their current performance and compare improvement packages within their budget. Image-assisted input will suggest visible features for user confirmation. Calculation engines and predictive models will produce the estimates; the AI assistant will explain these results and help users explore alternative scenarios.

03 High-level approach and architecture



Brief explanation: property information enters the backend, where scoring, retrofit and valuation models evaluate the selected scenario. The results are stored and presented through the dashboard with a plain-language AI explanation. A what-if request updates the scenario and reruns the relevant models.

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Proposed implementation and planned testing

03 Approach and architecture (continued)

Component	Planned role
Data preparation	Use synthetic or appropriately licensed public property, energy and sales data. Validate inputs and document missing readings and estimation assumptions.
Green Home Score	Calculate seven weighted component scores; show their contributions and identify the main causes of poor performance.
Retrofit planning	Recommend feasible measures such as insulation, heat pumps, glazing, solar and smart controls. Consider budget, prerequisites and interactions.
Impact estimation	Simulate each package sequentially. Report electricity and gas savings separately, convert them to carbon changes and estimate an indicative property-value range.
AI interaction	Use computer vision to assist property entry and a language model to explain computed results. Natural-language what-if questions trigger new calculations.

Scoring weights: energy rating 25%; energy consumption intensity 20%; insulation 15%; heating 15%; glazing 10%; renewable energy 10%; smart controls 5%. The weighted score is reported on a 0-100 scale with an efficiency classification.

Planned technology: React/TypeScript dashboard; Firebase authentication; Python/FastAPI backend; PostgreSQL storage; NetworkX retrofit planning; XGBoost valuation; PyTorch/YOLO and SAM for image assistance; a guarded language-model integration for explanations.

04 Planned testing methodology

Area	Method and acceptance focus
Data validation	Check input ranges, missing months and annualisation. Use reproducible train/validation/test splits by property, or by customer where records are linked, to prevent data leakage.
Scoring and retrofits	Test fixed weights, score boundaries, monotonic mappings and explanation of contributions. Check recommendation feasibility, ordering, budget constraints and interacting measures.
Energy and carbon	Test baseline-versus-scenario calculations for each fuel, including increased electricity after a heating change. Verify documented emission factors and consistent units.
Predictive models	Compare trained models against a simple baseline on held-out data. Report MAE/RMSE for numerical predictions, value-range coverage and recommendation ranking quality where reference labels exist.
AI and usability	Evaluate explanation accuracy and grounding in computed results; target ≥90% groundedness, ≥90% factual accuracy and <5% unsupported claims. Test image suggestions, missing-data cases and the full dashboard journey.

Demonstration: enter a sample home, inspect its score, select a feasible retrofit package and compare energy, carbon and indicative property-value results. Ask a what-if question and show the updated scenario. Report measured response times and evaluation outcomes as the prototype is completed.